

STRATEGIC USER OFFLOADING AND SERVICE PROVIDER PRICING IN MOBILE EDGE COMPUTING

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ABSTRACT

In mobile edge computing, users need to purchase computation and network resources from an Edge Service Provider (ESP) and a Network Service Provider (NSP) before offloading their tasks. The ESP and the NSP selfishly set their prices to maximize their revenues, while the users decide their offloading strategies based on the prices and compete for two resources. We study these strategic interactions as a two-stage Stackelberg game. We prove the existence of the equilibria for both the user offloading game and the service provider pricing game. To effectively compute these equilibria, we further design a greedy algorithm for the socially optimal user equilibrium, an incremental improvement algorithm for the service provider pricing equilibrium, and a branch-and-bound framework for the Stackelberg equilibrium. Simulations show that our approach closely approximates the oracle Stackelberg equilibrium.

Index Terms— Mobile Edge Computing, Task Offloading, Pricing Strategy, Strategic Interaction, Stackelberg Game

1. INTRODUCTION

Mobile Edge Computing (MEC) architecture [1] provides computation resources close to users and allows them to offload their delay-sensitive tasks to nearby edge servers. In practice, Edge Service Providers (ESPs) (e.g., Tencent Cloud and AWS) provide computation resources, and Network Service Providers (NSPs) (e.g., China Telecom and AT&T) provide network resources. To offload tasks successfully, users need to purchase both types of resources and inevitably compete for the limited computation and network resources with each other. Meanwhile, the ESPs and the NSPs are self-interested, setting prices to maximize their own revenues. The service providers' pricing schemes affect the users' resource demand, while the users' offloading decisions in turn influence the service providers' revenue-maximizing pricing. This motivates analyzing the strategic behaviours of the users and the service providers under separate management of the tightly coupled computation and network resources.

Extensive studies have been devoted to task offloading and pricing in MEC environments [1, 2, 3, 4, 5, 6, 7, 8, 9]. On the user side, existing studies typically focus on binary offloading decisions [1, 2] or on a single resource [3], and often overlook competition among users [4, 5]. On the service provider side, prior studies often assume a single-resource provider [6], centralized management of computation and network resources [7], or non-selfish service providers [8, 9]. Thus, there is no prior study investigating the users' and the service providers' strategic interactions when users compete for computation and network resources.

In this paper, we consider an ESP, an NSP, and a set of heterogeneous users, and formulate their strategic interactions as a two-stage Stackelberg Game. In Stage I, the ESP and the NSP strategically set their prices to maximize their revenues. In Stage II, users respond by deciding whether to offload their computation tasks and, if so, how much resource to acquire from each service provider. Such a game is highly complex with few existing solutions. Our main contributions are as follows: 1) We propose a novel game formulation by modeling the interactions among the ESP, the NSP, and users as a two-stage Stackelberg game. 2) We formulate the user offloading competition in Stage II as a Generalized Nash Equilibrium Problem (GNEP) [10, 11, 12], prove the existence of user equilibrium, and compute the socially optimal one. 3) We formulate the service provider pricing in Stage I as a pricing game with a fixed offloading user set. We show that the game has finite improvement property and thus admits a pricing equilibrium. We propose an incremental improvement algorithm to compute the equilibrium. 4) We propose a Branch-and-Bound (BnB) framework to efficiently identify the Stackelberg equilibrium by exploring different offloading user sets. Simulations show that our method effectively approximates the oracle Stackelberg equilibrium.

2. SYSTEM MODEL AND STACKELBERG GAME FORMULATION

In this section, we first introduce the system model, and then formulate the two-stage Stackelberg game.

2.1. System Model.

We consider a MEC environment with an ESP, an NSP, and a set of users $\mathcal{I} = \{1, \dots, I\}$. The users can execute tasks on their local devices or offload them to the ESP's edge server via the NSP's network. We begin by presenting the user model, followed by the service provider model.

The User Model. Each user $i \in \mathcal{I}$ has an indivisible task characterized by workload w_i (in CPU cycles) and data size d_i (in Bytes). The user i makes a binary offloading decision $o_i \in \{0, 1\}$, where $o_i = 1$ corresponds to offloading and $o_i = 0$ to local computation. If it chooses to offload, it also needs to decide the amount of computation resource f_i (in CPU cycles/s) and bandwidth b_i (in Hz) to purchase from the ESP and the NSP, respectively.

1) *The Task Delay.* If user $i \in \mathcal{I}$ finishes its task with its local computation resource f_i^l (in CPU cycles/s), the delay of local computation is $T_i^l = w_i / f_i^l$. If it offloads its task to the ESP's edge server, the delay of computation offloading consists of transmission delay and computation delay. The transmission delay of user i 's task is $T_i^u(b_i) = d_i / (b_i \sigma_i)$, where σ_i is the channel spectral efficiency (in bits/s/Hz) between user i and the NSP. We ignore the downlink transmission as the size of the task result is negligible compared to the size of the input [1]. The computation delay at the ESP's edge server is $T^c(f_i) = w_i / f_i$. The total delay of computation offloading is

$$T_i^e(f_i, b_i) = T_i^u(b_i) + T_i^c(f_i). \quad (1)$$

2) *The User's Costs.* Each user $i \in \mathcal{I}$ has a delay sensitivity α_i , which reflects the marginal monetary cost incurred by the task delay. The user i 's local computation cost is $C_i^l = \alpha_i T_i^l$. His computation offloading cost includes the delay cost and payments to the ESP and NSP:

$$C_i^e(f_i, b_i) = \alpha_i T_i^e(f_i, b_i) + p_E f_i + p_N b_i, \quad (2)$$

where p_E and p_N are the unit prices of computation resource and bandwidth, respectively. The overall cost of user i is:

$$C_i(o_i, f_i, b_i) = o_i C_i^e(f_i, b_i) + (1 - o_i) C_i^l. \quad (3)$$

The ESP Model. The ESP manages an edge server with limited computation resource F . The total resource usage should not exceed the capacity, i.e., $\sum_{i \in \mathcal{I}} f_i \leq F$. The ESP incurs a unit cost $c_E > 0$ for computation resource, and charges users a unit price $p_E \geq c_E$ to ensure economic rationality. The ESP's revenue is $U_E = (p_E - c_E) \sum_{i \in \mathcal{I}} f_i$.

The NSP Model. The NSP manages a base station with a total wireless bandwidth B . The total bandwidth usage should not exceed the capacity, i.e., $\sum_{i \in \mathcal{I}} b_i \leq B$. Similarly, the unit cost and unit price of bandwidth are denoted by c_N and p_N , respectively. The NSP's revenue is $U_N = (p_N - c_N) \sum_{i \in \mathcal{I}} b_i$.

2.2. Stackelberg Game Formulation

We model the interactions between the users and two service providers as a two-stage Stackelberg game. In Stage I, the

ESP and the NSP set the prices of computation and network resources, respectively. In Stage II, given these prices, the users make offloading decisions to minimize their costs.

Stage I: Service Provider Pricing Game. The ESP and the NSP selfishly set their prices p_E and p_N to maximize their revenues. We highlight that each user i 's resource usage f_i and b_i depend on the prices p_E and p_N , and thus we denote them explicitly as $f_i(p_E, p_N)$ and $b_i(p_E, p_N)$. The ESP solves the following revenue maximization problem:

$$\max_{p_E} U_E(p_E, p_N) = (p_E - c_E) \sum_{i \in \mathcal{I}} f_i(p_E, p_N) \quad (4a)$$

$$s.t. \quad p_E \geq c_E, \quad \sum_{i \in \mathcal{I}} f_i(p_E, p_N) \leq F. \quad (4b)$$

Similarly, the NSP solves the following problem:

$$\max_{p_N} U_N(p_N, p_E) = (p_N - c_N) \sum_{i \in \mathcal{I}} b_i(p_E, p_N) \quad (5a)$$

$$s.t. \quad p_N \geq c_N, \quad \sum_{i \in \mathcal{I}} b_i(p_E, p_N) \leq B. \quad (5b)$$

Stage II: User Offloading Game. We formulate the user offloading game as a GNEP. Each user $i \in \mathcal{I}$ selects offloading decision o_i and resource usage (f_i, b_i) to minimize cost:

$$\min_{o_i, f_i, b_i} C_i(o_i, f_i, b_i) = o_i C_i^e(f_i, b_i) + (1 - o_i) C_i^l \quad (6a)$$

$$s.t. \quad o_i \in \{0, 1\}, f_i \geq 0, b_i \geq 0, \quad (6b)$$

$$f_i + \sum_{j \in \mathcal{I}, j \neq i} f_j \leq F, b_i + \sum_{j \in \mathcal{I}, j \neq i} b_j \leq B. \quad (6c)$$

The constraint (6c) is the system-wide resource limits imposed by the ESP and NSP. It implies a coupled strategy space where each user's strategy space depends on other users' strategies [10].

Stackelberg Equilibrium. We define $s_i = (o_i, f_i, b_i)$, $\mathbf{s} = (s_1, \dots, s_I)$, and let $\mathbf{s}_{-i}$ denote the strategy profile of all users except the user i . Given a fixed $\mathbf{s}_{-i}$, we denote the strategy space of user i defined by (6b), (6c) as $\mathcal{S}_i(\mathbf{s}_{-i})$. And we denote the joint strategy space of all users as $\mathcal{S}$. We are interested in deriving the Stackelberg equilibrium (Definition 1) of the Stackelberg game, where no party can be better off by unilateral deviation. Thus, the Stackelberg equilibrium represents a stable outcome of the strategic interactions between users and service providers.

Definition 1. The strategy profile $(p_E^*, p_N^*, \mathbf{s}^*)$ is a Stackelberg equilibrium if the following conditions hold,

$$U_E(p_E^*; p_N^*) \geq U_E(p_E; p_N^*), \forall p_E, \quad (7)$$

$$U_N(p_N^*; p_E^*) \geq U_N(p_N; p_E^*), \forall p_N, \quad (8)$$

$$C_i(s_i^*) \leq C_i(s_i), \forall s_i \in \mathcal{S}_i(\mathbf{s}_{-i}^*), \forall i \in \mathcal{I}. \quad (9)$$

We use backward induction to derive the Stackelberg equilibrium by first solving the user offloading game in Stage II and then the service provider pricing game in Stage I.

3. STAGE II: USER OFFLOADING GAME AND SOCIALLY OPTIMAL EQUILIBRIUM

In this section, we study the user offloading game given service providers' prices. We first establish the existence of the user equilibrium, and then compute the socially optimal one.

A feasible strategy profile $\mathbf{s}^* \in \mathcal{S}$ is a Generalized Nash Equilibrium (GNE) [10] of the user offloading game if no user can reduce their cost by unilateral deviation. We show that the user offloading game is an exact generalized potential game [11, 13], since it admits an exact potential function $\Phi(\mathbf{s}) = \sum_{i \in \mathcal{I}} C_i(s_i)$ such that for every $i \in \mathcal{I}$ and all feasible strategy pairs $(s_i, \mathbf{s}_{-i}), (s'_i, \mathbf{s}_{-i}) \in \mathcal{S}$, $C_i(s'_i, \mathbf{s}_{-i}) - C_i(s_i, \mathbf{s}_{-i}) = \Phi(s'_i, \mathbf{s}_{-i}) - \Phi(s_i, \mathbf{s}_{-i})$ holds. As every potential game admits a Nash equilibrium [14], the existence of user equilibrium is established as follows.

Theorem 1. *The user offloading game has at least one GNE.*

To characterize users' offloading behavior, we first derive lower and upper bounds on their resource usage. If a rational user $i \in \mathcal{I}$ chooses to offload, it would select a resource usage (f_i, b_i) such that the offloading cost is smaller than the local computation cost, i.e., $C_i^e(f_i, b_i) \leq C_i^l$. For each user $i \in \mathcal{I}$, the offloading cost $C_i^e(f_i, b_i)$ is strictly convex in f_i and b_i , and thus admits a unique minimizer $(\hat{f}_i, \hat{b}_i) \triangleq (\sqrt{\alpha_i w_i / p_E}, \sqrt{\alpha_i d_i / \sigma_i p_N})$. Define thresholds $f_i^{\text{thres}}, b_i^{\text{thres}}$ such that $C_i^e(f_i^{\text{thres}}, \hat{b}_i) = C_i^l$ and $C_i^e(\hat{f}_i, b_i^{\text{thres}}) = C_i^l$. Hence, we establish the following lemma.

Lemma 1. *A rational offloading user i 's resource usage (f_i, b_i) satisfies $f_i^{\text{thres}} \leq f_i \leq \hat{f}_i$ and $b_i^{\text{thres}} \leq b_i \leq \hat{b}_i$.*

Any resource usage outside this region is dominated, as exceeding $\hat{f}_i$ or $\hat{b}_i$ yields a higher offloading cost than $C_i^e(\hat{f}_i, \hat{b}_i)$, and falling below f_i^{thres} or b_i^{thres} makes offloading more costly than local computation.

We compute the socially optimal user equilibrium by solving the following Social Cost Minimization (SCM) problem:

$$\text{SCM} : \min_{\mathcal{X} \subseteq \mathcal{I}} \min_{(f_i, b_i)_{i \in \mathcal{X}}} \sum_{i \in \mathcal{X}} C_i^e(f_i, b_i) + \sum_{j \in \mathcal{I} \setminus \mathcal{X}} C_j^l \quad (10a)$$

$$\text{s.t. } C_i^e(f_i, b_i) \leq C_i^l, \forall i \in \mathcal{X}, \quad (10b)$$

$$f_i^{\text{thres}} \leq f_i \leq \hat{f}_i, b_i^{\text{thres}} \leq b_i \leq \hat{b}_i, \forall i \in \mathcal{X}, \quad (10c)$$

$$\sum_{i \in \mathcal{X}} f_i \leq F, \sum_{i \in \mathcal{X}} b_i \leq B. \quad (10d)$$

The SCM problem is a two-level problem: the outer problem selects the offloading user set that minimizes the overall social cost, while the inner problem allocates resources among a given offloading user set $\mathcal{X}$.

1) *Optimal Resource Allocation in the Inner Problem.*

The inner problem is strictly convex and solvable using standard methods (e.g., primal-dual decomposition). Given a

non-empty offloading user set $\mathcal{X}$, define $\hat{F}_{\mathcal{X}} \triangleq \sum_{i \in \mathcal{X}} \hat{f}_i$ and $\hat{B}_{\mathcal{X}} \triangleq \sum_{i \in \mathcal{X}} \hat{b}_i$. We establish the following proposition based on the KKT conditions.

Proposition 1. *Given a non-empty offloading user set $\mathcal{X}$, the optimal solution $(f_i^*, b_i^*)_{i \in \mathcal{X}}$ to the inner problem falls into one of the following four cases:*

$$\begin{cases} f_i^* = \hat{f}_i, b_i^* = \hat{b}_i, \forall i \in \mathcal{X}, & \text{if } \hat{F}_{\mathcal{X}} \leq F, \hat{B}_{\mathcal{X}} \leq B, \\ \sum_{i \in \mathcal{X}} f_i^* = F, b_i^* = \hat{b}_i, \forall i \in \mathcal{X}, & \text{if } \hat{F}_{\mathcal{X}} > F, \hat{B}_{\mathcal{X}} \leq B, \\ f_i^* = \hat{f}_i, \forall i \in \mathcal{X}, \sum_{i \in \mathcal{X}} b_i^* = B, & \text{if } \hat{F}_{\mathcal{X}} \leq F, \hat{B}_{\mathcal{X}} > B, \\ \sum_{i \in \mathcal{X}} f_i^* = F, \sum_{i \in \mathcal{X}} b_i^* = B, & \text{if } \hat{F}_{\mathcal{X}} > F, \hat{B}_{\mathcal{X}} > B. \end{cases}$$

When the aggregate ideal resource demands $(\hat{F}_{\mathcal{X}}, \hat{B}_{\mathcal{X}})$ exceed the resource capacities, the optimal solution must exhaust the limited resources; otherwise, each user's optimal usage equals its ideal amount. This proposition will be useful for subsequent analysis of service providers' pricing in Section 4.

2) *Optimal Offloading User Set Selection in the Outer Problem.* We prove that the outer problem is NP-hard via a reduction from the 0-1 knapsack problem.

Proposition 2. *The outer problem is NP-hard.*

We propose a greedy algorithm to approximate the minimum social cost. The algorithm maintains an offloading user set $\mathcal{X}$. Define $V(\mathcal{X})$ as the social cost with offloading user set $\mathcal{X}$. Suppose user $j \notin \mathcal{X}$ provides the largest marginal reduction in social cost $V(\mathcal{X}) - V(\mathcal{X} \cup \{j\})$. If the marginal reduction is positive, the algorithm will add the user j to set $\mathcal{X}$, and repeat such a process until no remaining user provides a positive marginal reduction in social cost.

4. STAGE I: SERVICE PROVIDER PRICING GAME AND STACKELBERG EQUILIBRIUM COMPUTATION

In this section, we first consider a restricted pricing game between service providers and compute the pricing equilibrium via an incremental improvement algorithm. Then, we propose a BnB framework to compute the Stackelberg equilibrium.

Given a fixed offloading user set $\mathcal{X}$, the ESP and the NSP engage in a restricted pricing game. Its joint pricing space $\mathcal{P}_{\mathcal{X}}$ consists of price pairs (p_E, p_N) where users admit an equilibrium with offloading user set $\mathcal{X}$. The pricing equilibrium is a price pair $(p_{E, \mathcal{X}}^*, p_{N, \mathcal{X}}^*) \in \mathcal{P}_{\mathcal{X}}$ where neither service provider can improve its revenue by unilateral deviation.

We prove the existence of pricing equilibrium by showing that the pricing game satisfies the finite improvement property [14], which is based on the following two properties.

1) *Monotonicity of Service Provider Revenue.* Recall that the service providers' revenues U_E and U_N depend on offloading users' aggregate demands $\sum_{i \in \mathcal{X}} f_i$ and $\sum_{i \in \mathcal{X}} b_i$.

According to Proposition 1, these demands at socially optimal user equilibrium are piecewise functions of p_E and p_N :

$$\sum_{i \in \mathcal{X}} f_i^* = \begin{cases} \sum_{i \in \mathcal{X}} \sqrt{\frac{\alpha_i w_i}{p_E}}, & \text{if } p_E > \left(\frac{\sum_{i \in \mathcal{X}} \sqrt{\alpha_i w_i}}{F} \right)^2, \\ F, & \text{if } p_E \leq \left(\frac{\sum_{i \in \mathcal{X}} \sqrt{\alpha_i w_i}}{F} \right)^2, \end{cases} \quad (11)$$

$$\sum_{i \in \mathcal{X}} b_i^* = \begin{cases} \sum_{i \in \mathcal{X}} \sqrt{\frac{\alpha_i d_i}{\sigma_i p_N}}, & \text{if } p_N > \left(\frac{\sum_{i \in \mathcal{X}} \sqrt{\alpha_i d_i / \sigma_i}}{B} \right)^2, \\ B, & \text{if } p_N \leq \left(\frac{\sum_{i \in \mathcal{X}} \sqrt{\alpha_i d_i / \sigma_i}}{B} \right)^2. \end{cases} \quad (12)$$

By substituting Eq.(11) and Eq.(12) into U_E and U_N , we have the following lemma.

Lemma 2. *The service providers' revenues U_E and U_N are continuous and monotonically increasing in p_E and p_N , respectively.*

2) *Closedness and Boundedness of Joint Pricing Space.* To show that the joint pricing space $\mathcal{P}_{\mathcal{X}}$ is closed and bounded, we construct its superset $\mathcal{P}_{\mathcal{X}}^{\text{pre}} \triangleq \{(p_E, p_N) : C_i^e(\hat{f}_i, \hat{b}_i) \leq C_i^l, \forall i \in \mathcal{X}, \sum_{i \in \mathcal{X}} f_i^{\text{thres}} \leq F, \sum_{i \in \mathcal{X}} b_i^{\text{thres}} \leq B\}$ using necessary conditions of user equilibrium. Specifically, each user i needs to satisfy $C_i^e \leq C_i^l$ at its minimum cost, and the aggregate threshold demands do not exceed resource capacity. The set $\mathcal{P}_{\mathcal{X}}^{\text{pre}}$ is closed and bounded due to the continuity of its constraints over a finite domain. Hence, we have the following lemma.

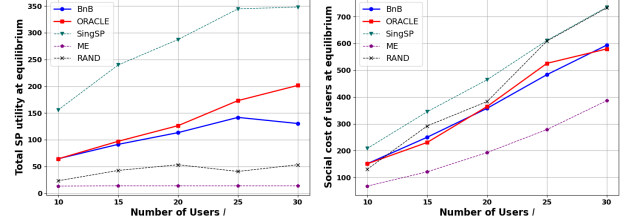
Lemma 3. *Set $\mathcal{P}_{\mathcal{X}}$ is bounded since $\mathcal{P}_{\mathcal{X}} \subseteq \mathcal{P}_{\mathcal{X}}^{\text{pre}}$. It is closed since any convergent sequence in $\mathcal{P}_{\mathcal{X}}$ has a limit that leads to a user equilibrium with offloading user set $\mathcal{X}$.*

We can establish the finite improvement property based on Lemma 2 and Lemma 3, i.e., any revenue improvement path is finite. Therefore, we have the following theorem.

Theorem 2. *Given any non-empty offloading user set $\mathcal{X}$, the service provider pricing game admits at least one pricing equilibrium.*

We propose an incremental improvement algorithm to compute the pricing equilibrium given a fixed offloader set $\mathcal{X}$. In this algorithm, the ESP and NSP start from an initial point in $\mathcal{P}_{\mathcal{X}}$, and iteratively increase their prices until the prices reach the boundary of $\mathcal{P}_{\mathcal{X}}$.

To compute the Stackelberg equilibrium, we look for the offloading user set $\mathcal{X}^*$ that yields the maximum total revenue for the ESP and the NSP at the corresponding pricing equilibrium. We propose a BnB framework to search over all possible offloading user sets. It branches on candidate offloading user sets and prunes suboptimal ones using upper bounds on the service providers' revenue. For each offloading user set $\mathcal{X}$, the ESP and NSP reach a pricing equilibrium $(p_{E,\mathcal{X}}^*, p_{N,\mathcal{X}}^*)$ and evaluate their total revenue $U_E(p_{E,\mathcal{X}}^*, p_{N,\mathcal{X}}^*) + U_N(p_{E,\mathcal{X}}^*, p_{N,\mathcal{X}}^*)$. We have the following theorem.



(a) Total revenue of ESP and NSP. (b) Social cost of users.

Fig. 1. Comparison of our method and baselines with different numbers of users.

Theorem 3. *The pricing equilibrium $(p_{E,\mathcal{X}^*}^*, p_{N,\mathcal{X}^*}^*)$ and the associated socially optimal user equilibrium with offloading user set $\mathcal{X}^*$ constitute a Stackelberg equilibrium.*

5. NUMERICAL RESULTS

We compare our proposed method (BnB) with four baseline methods: Oracle, Market Equilibrium (ME) [8], SingleSP [7], and Random Offloading (RAND). The Oracle baseline finds the Stackelberg equilibrium via brute-force search. The ME baseline identifies the market equilibrium, representing non-strategic service providers with competitive users. SingleSP assumes a single selfish service provider managing both resources, with non-competitive users simply following its instructions to offload. RAND randomly selects offloading users and computes the corresponding pricing equilibrium, capturing strategic service providers with non-strategic users.

As shown in Fig. 1, our method closely approximates the Oracle solution, achieving 86.0% of service provider revenue and 95.8% of user social cost. The Stackelberg-based models (Oracle, BnB) yield higher service provider revenue than ME and RAND but lower than SingleSP (Fig.1(a)), while delivering lower user cost than SingleSP and RAND but higher than ME (Fig.1(b)). This indicates that (i) selfish pricing between the ESP and the NSP improves their total revenue over non-strategic provider models (e.g., ME) but remains below centralized pricing (e.g., SingleSP), and (ii) selfish user competition reduces user costs compared to non-competitive users (e.g., SingleSP and RAND), though costs rise compared to ME due to the service providers' selfish pricing.

6. CONCLUSION

We study user-service provider strategic interactions in MEC via a Stackelberg game. We prove the existence of equilibria in both the user offloading game and the service provider pricing game. We propose a greedy algorithm for the socially optimal user equilibrium, an incremental improvement algorithm for the pricing equilibrium, and a BnB framework for the Stackelberg equilibrium. Experiments show that our method closely approximates the oracle Stackelberg equilibrium. Future work will extend to multiple ESPs and NSPs.

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